



Client-Tracking in IOS-XE

There are three different ways for Meraki devices to identify clients: **Unique client identifier**, **Track by MAC**, and **Track by IP**. These tracking methods are how key information like the clients list and network usage data is populated in the dashboard. The MX Security Appliance has the option to use any of the 3 options with details being found in [Client-Tracking Options](#). Onboarding a Catalyst 9000 series switch will automatically enforce "Unique Client Identifier" as the tracking method for the network.



Note: The following ports don't support Client Tracking features on MS390 & C9300/L/X-M.

1. Ports with supported max speeds 25G, 40G, 100G.
2. Link Aggregation ports

Unique Client Identifier

Unique client identifier is a Meraki technology that leverages network topology and device information to uniquely identify and track clients. It uses an algorithm that intelligently correlates client MAC and IP addresses seen across the Meraki stack, allowing the security appliance to generate a unique identifier for each client in a [combined network](#) with other Meraki devices. This is specifically useful when there are Meraki MS switches routing layer 3 between end clients and the security appliance, which segregates broadcast traffic containing the client's MAC address.

Tracking by unique client identifier also disables uplink sampling for clients, which can be helpful in certain scenarios where non-Meraki NAC solutions are deployed in mixed vendor environments.



Note: Some tools, such as client connectivity alerts and client ping, are based on ARP and will not be available when using Unique client identifier.

Requirements and Conditions

Please review the requirements and conditions below before enabling this feature on your network.

To see the Unique Client Identifier option in Addressing & VLANs, the following conditions must be met:

- There must be a security appliance with at least one Meraki MS or Meraki Managed/Monitored Catalyst L3 switch in the same network in the dashboard. To avoid incorrect tracking data, the devices in this dashboard network should also be in the same physical network.
- This option is only shown if the MX firmware version is 9+ and the MS firmware version is 10+.
- Do not use Unique Client Identifier in a dashboard network where the MX's WAN ports are connected to a Meraki switch in the same Dashboard network. If you need to use a Meraki switch in between your ISP and the MX WAN please isolate this switch into a separate Dashboard network.



Note: When modifying the 'Client tracking' the change will reset any client device with a manually configured group policy associated. Manual group

 policies are on the Network-Wide > Monitor > Clients page under the policy column. If a policy is needed for a particular associated device, it must be re-added once the change is made and the device populates on the client list.

Search Dashboard

+

Map

-

Satellite

Address, zip code, etc.

Go

O'Farrell St

O'Farrell St

Divisadero St

Turk Blvd

Divisadero St

Golden Gate Ave

Anzavista Ave

Fortuna Ave

Networks

Network tags

Tag

Combine

Delete

Search...

13 networks

Over the last week: 29 clients, 1 SM device, 102.10 GB

☐

☐

☐

☐

☒

☒

☒

☐

☐

☐

☐

☐

☐

☐

☐

Dashboard now allows MR, MS, MX, MC, MV, and Systems Manager products to all be managed in a single network, showing wired and wireless clients together.

To help transition, this tool allows you to take existing, separate networks and combine them into a single network. When combining, all of the configuration and monitoring data of the separate networks is kept.

This tool can only combine networks of different types. You cannot, for example, combine two wireless networks together. In that case, move the APs to the desired network, and delete the old network.

When combining an MR network with an MX with wireless capabilities, or Z-Series device, be aware that the security appliance or teleworker gateway will not mesh with the MR access point. Some aspects of the client experience, such as roaming, may be negatively affected by this network configuration.

When combining networks that have differing beta firmware preferences, the settings will be uniformly applied across the networks, with a default of not trying beta firmware. For example, if all of the networks are configured to try beta firmware, they will continue to have this setting. If at least one of the networks is set to not try beta firmware, then all of the other networks will be set as such. Once combined, you can toggle your beta preferences by visiting Configure->General.

For more information, visit the [Combined Networks](#) knowledge base article.

Network name:

Networks to be combined:
main - appliance
main - switch
main - wireless

Security appliance networks tracking clients by IP addresses cannot be combined with other networks

Combine

Tags

Network type

Combined

MDM

MDM

Appliance

Switch

Wireless

Combined

Combined

Combined

Camera

Switch

Switch

Appliance

Cloud Monitoring

Information further detailing Cloud Monitoring can be found in [Cloud Monitoring for Catalyst Onboarding](#) and [Cloud Monitoring Required Configuration](#).

During the onboarding of a Cloud Monitored Catalyst 9000 switch, after dashboard communication is established, dashboard will access the device via the secure TLS tunnel using the meraki-user account over SSH and apply a policy configuration via NETCONF to collect client data such as IP Address and MAC Address.

Device Tracking Policy "MERAKI_POLICY"

Policy application

The device tracking policies named MERAKI_POLICY is added to each Layer 2 interface at the time of onboarding **except** the following:

- Detected uplink interface
- Interfaces known to connect directly to other devices in your dashboard network (including both Catalyst and Meraki hardware)
- Device tracking explicitly disabled
- If an interface has device-tracking policy already applied, then dashboard will not attach MERAKI_POLICY
- In case of port-channel, the port channel will receive MERAKI_POLICY and the member interfaces of a port channel will not.
- SVL interfaces



MERAKI_POLICY is applied to all eligible interfaces:
device-tracking policy MERAKI_POLICY
security-level glean
no protocol udp
tracking enable

Policy re-application

If the dashboard configuration that is applied to an interface is modified, Dashboard will change the configuration back to the required configuration to allow for clients to be gleaned in **some** instances;

- If a configuration was applied by dashboard, such as *Device Tracking & TA flow monitor*, any user's changes will be **re-applied** back to cloud expected configuration
- If a configuration was not applied by dashboard, then dashboard will **not** modify, edit or re-apply any changes that are made locally on the device.

Disable Device Tracking

Pre-Onboarding

- To disable device tracking on interfaces **before** onboarding, please create and apply the following policy on the desired interfaces before connecting the switch to the Meraki Dashboard via the onboarding application:

```
device-tracking policy NO_TRACK
  security-level glean
  no protocol ndp
  no protocol dhcp6
  no protocol arp
  no protocol dhcp4
  no protocol udp
exit
interface <Interface to modify>
  device-tracking policy NO_TRACK
end
```

Post-Onboarding

- Disabling device tracking on interfaces **after** the onboarding can only be performed device-wide. To do so, please edit the existing MERAKI_POLICY, by replacing the tracking settings with the configuration used in the above NO_TRACK policy example:

```
device-tracking policy MERAKI_POLICY
  no protocol ndp
  no protocol dhcp6
  no protocol arp
  no protocol dhcp4
  no protocol udp
end
```

Cloud Management: Device Configuration

Information further detailing Cloud Management can be found in [Cloud Management with IOS XE Overview](#)

During the onboarding of a Cloud Managed: Device Configuration 9000 switch, CLI configuration will be applied to the device to collect client data such as IP Address and MAC Address. There are 3 types of policies that can be applied, Access, Trunk and Uplink.



Note: During onboarding of a C9500 to Device Configuration mode, device tracking will not be modified. Users can choose to modify it via CLI after onboarding.

Device Tracking Policy

Access Ports

```
device-tracking policy MERAKI_ACCESS_TRACK
  limit address-count 1000
  security-level glean
  tracking enable
```

Uplink

```
device-tracking policy MERAKI_NO_TRACK
  trusted-port
  security-level glean
```

```
no protocol ndp
no protocol dhcp6
no protocol arp
no protocol dhcp4
```

Trunk

```
device-tracking policy MERAKI_TRUNK_TRACK
limit address-count 32000
security-level glean
```



Note: During the onboarding, the above tracking policies will only be applied to interfaces without existing device tracking configuration.

Disable Device Tracking

Device tracking can be disabled either before On-boarding at an **interface level** or after On-boarding at a **switch level**. To disable device tracking on an interface level for switches that are already On-boarded, the device must be Off-boarded, follow the Pre-Onboarding steps and On-board the switch again.

Pre-Onboarding

- To disable device tracking on interfaces **before** on-boarding, please create and apply the following policy on the desired interfaces before connecting the switch to the Meraki Dashboard via the onboarding application:

```
device-tracking policy NO_TRACK
no protocol ndp
no protocol dhcp6
no protocol arp
no protocol dhcp4
no protocol udp
exit
interface <Interface to modify>
device-tracking policy NO_TRACK
end
```

Post-Onboarding

- Disabling device tracking on interfaces **after** the on-boarding can only be performed device wide. To do so, please edit the existing policy:

```
device-tracking policy <POLICY_NAME>
no protocol ndp
no protocol dhcp6
no protocol arp
no protocol dhcp4
no protocol udp
exit
```



Note: When the Meraki device-tracking policy is removed from the interface after the onboarding it will be re-applied automatically. If the policy is replaced with a custom (non-Meraki) policy it will not be overwritten.

Additional Resources

- [Renaming Clients in Dashboard](#)
- [Client Details Page Overview](#)
- [Client Usage Page Overview](#)